NV 1047

256/263/300 Channel TFT LCD Gate Driver IC

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Information is current as of publication date. It is subject to change without notice.

July. 2011 Ver. 0.3

Features

- Gate driver LSI for active-matrix LCD
- Gate driver outputs: 256/263/300
- Each output channel can output two levels.
- Bidirectional shift controllable and cascadeable.
- Maximum gate-on voltage: +35 V
- Maximum gate-off voltage: -15 V
- Maximum operating frequency: 200 KHz
- Logic input voltage:2.2 ~ 3.6 V
- High voltage CMOS process
- Fine pitch TCP/COF packages

Description

The NV1047 is a 256/263/300-channel gate driver suitable for XGA, SXGA and QXGA size TFT liquid crystal display panels. It takes in a start up pulse and generates 256/263/300 scanning pulses to be applied to LCD's gate electrodes. In combining with suitable LCD source drivers, display data are written into LCD panel.

NV1047 is a high voltage CMOS IC. A wide range of settings are available, with a maximum gate-on voltage of +35V and a gate-off voltage of –15V for output channels. The input and control line voltage level can be operated from 2.2V to 3.6V. The operating frequency of the pulse scanning is up to 200 kHz, enough for QXGA display panels. It also has a chip level enable function to turn on and off all the output channels at once.

NV1047 can be shipped in Chip-on-Film (COF) packages. Detail tape requirement can be inquired to New Vision Microelectronics Inc.

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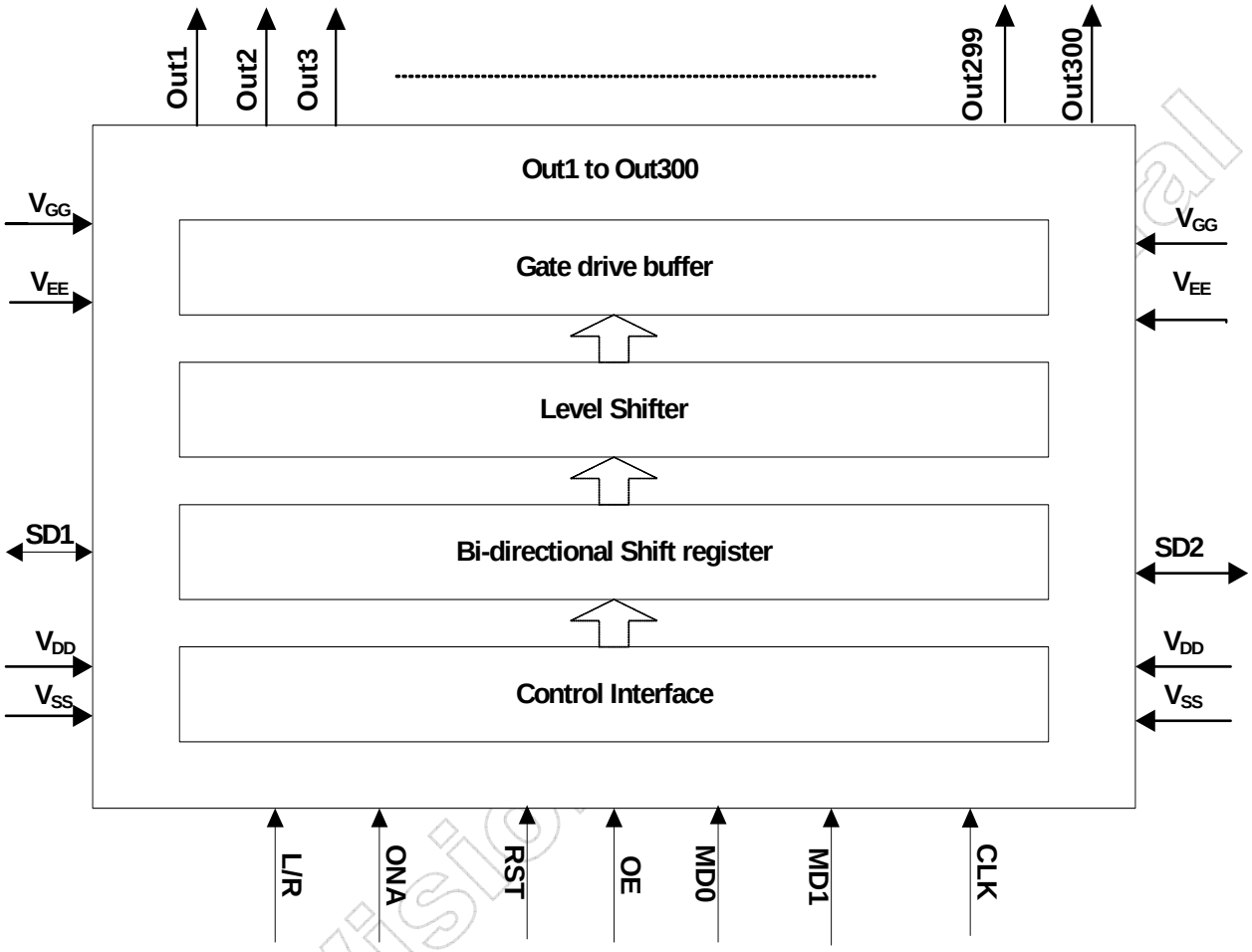
Block Diagram



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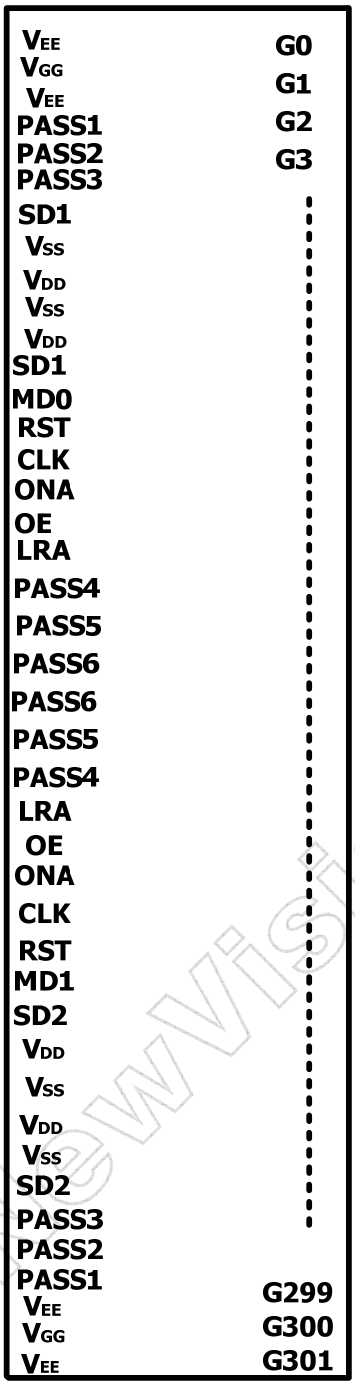
Pin Arrangement



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Pin Function

Pin Name	Quantity	I/O	Functions
V_{GG}	2	Power	Positive power supply for the output driver. Maximum: 35V
V_{EE}	4	Power	Negative power supply for the output driver. Minimum: -15V.
V_{DD}	4	Power	Positive power supply for the low-voltage CMOS logic circuit. Typical: 3.3V
Vss	4	Power	Negative power supply for the low-voltage CMOS logic circuit. Typical: 0V
SD1 SD2	2 2	I/O	Shift data input/output pins. These pins are used to input/output data to/from shift register chains. When used as an input, data are captured in sync with the rising edge of CLK. When used as an output, data are output in sync with the falling edge of CLK. When L/R = "H", SD1 is input and SD2 is output. When L/R = "L" (as Power-up default), SD2 is input and SD1 is output.
CLK	2	I	Master clock for internal shift register. Data are shifted in sync with the rising edge of this pin.
ONA	2	I	Output all-on control pin. Power-up default is "H". When this pin is active ("L"), all the output pins will be forced to V_{GG} level. This pin does not clear data in shift registers. This signal is asynchronous to CLK.
OE	2	I	Channel output control pin. Power-up default is "L". When this pin is active ("H"), the output level will be set at V_{EE}. This pin does not clear data in shift registers. This signal is asynchronous to CLK.
L/R	2	I	Shift direction control pin to determine upward scanning or downward scanning When L/R = "V_{DD}"(default) SD1 → Out1 → Out2 ---- Out299 → Out300 → SD2 When L/R = "V_{SS}" SD2 → Out300 → Out299 ---- Out2 → Out1 → SD1
Out[1:300]	300	O	Gate drive outputs. The high level is V_{GG} and low level is V_{EE} depending on the data stored in shift register and the state of OE. The Out0-0 and Out-301 are dummy pins.
MD[1:0]	2	I	Gate channel Shift Mode pins MD[1:0]=11, 300 Gate channel, Out[1:300] valid; **(Default)** MD[1:0]=10, 263 Gate channel, Out[1:135], Out[166:300] valid; MD[1:0]=01, 256 Gate channel, Out[1:128], Out[173:300] valid; MD[1:0]=00, Not support
PASS	2 × 6	Internal Pass	Internal Pass 6 pairs of pads are shorted internally.

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Function Description

Data Shift Register

When L/R="H", the SD1 start pulse input is sensed at the rising edge of CLK and latched into the first stage of the shift register. It also results in the first scan signal to be output from the Out1 output pin. At the rising edge of next CLK pulse, the latched data is transferred to the next stage of the shift register and the new data of SD1 is sensed and latched simultaneously.

So by providing a one-clock width pulse at the SD1 pin, the pulse is shifted down towards Out300 and SD2 at every rising edge of the CLK pulse. When L/R="L", the pulse shifting direction is from SD2 to SD1. The following table summarizes the shifting operation:

MD[1:0]	channel	Shift direction L/R=H
01	256	SD1→Out1→Out2→…→Out128→Out173→…→Out300→SD2
10	263	SD1→Out1→Out2→…→Out132→Out170→…→Out300→SD2
11	300	SD1→Out1→Out2→Out3→…→Out298→Out299→Out300→SD2

MD[1:0]	channel	Shift direction L/R=L
01	256	SD1←Out1←Out2←…←Out128←Out173←…←Out300←SD2
10	263	SD1←Out1←Out2←…←Out132←Out170←…←Out300←SD2
11	300	SD1←Out1←OUT2←Out3←…←Out298←Out299←Out300←SD2

Gate Output

The output pin (Out1 thru Out300) provides V voltage or V_{EE} voltage to the TFT-LCD panel depending on the data stored in the shift register. For normal operation, a V_{GG} voltage is output one by one from Out1 to Out300 in sync with CLK pulse. Out0 and Out301 are always fixed to V_{EE} When L/R="H", after 300 CLK rising edges are completed, the SD2 goes up to the HIGH level at the 300th falling edge of CLK and goes down to LOW level at the 301st falling edge of CLK pulse. This SD2 output signal becomes the SD1 start pulse input of the next gate driver IC in cascade.

The gate output voltage levels are illustrated in the following figure.

When OE pin is set "H", all gate output channels are forced to V_{EE} regardless of CLK. The channel output returns to normal status as soon as OE_ pin goes back to "L". OE_ does not influence the shift register and is asynchronous to CLK.

All gate output channels can be switched to V_{GG} by setting ONA_ = "L". ONA_ pin is also asynchronous to CLK. The channel outputs return to the normal status as soon as ONA_ pin goes back to "H". ONA_ does not influence the shift register.

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Electric Specification

Absolute maximum rating (referenced to T_A = 25°C, V_{SS} = 0 V)

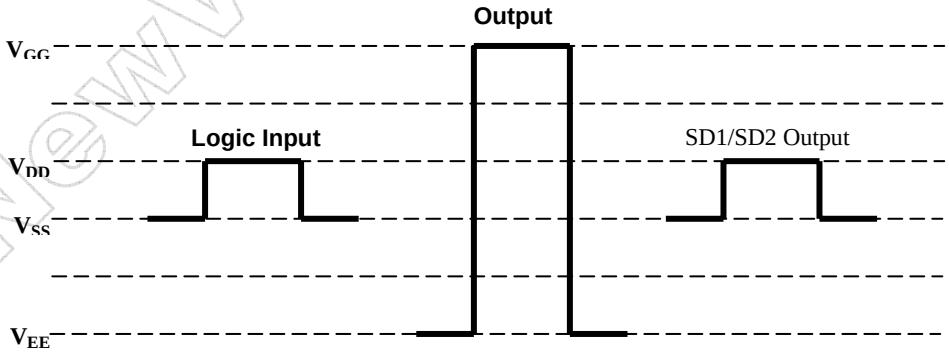
Item	Symbol	Specification	Unit
Power supply (1)	V_{DD}	–0.3 to + 7.0	Volts
Power supply (2)	V_{GG}	–0.3 to + 40.0	Volts
Power supply (3)	V_{EE}	–22.0 to + 0.3	Volts
Power supply (4)	V_{GG} - V_{EE}	–0.3 to + 48.0	Volts
Input voltage	V_{IN}	–0.3 to V_{DD} + 0.3	Volts
Storage temperature	T_{STORE}	–55 to +125	°C

Stresses beyond those listed under maximum ratings may cause permanent damage to the device.
Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability (see recommended operating conditions).
Maintain the following voltage relations:
$V_{EE} \leq V_{SS} \leq V_{GG}$
$V_{EE} \leq V_{SS} \leq V_{DD}$
Recommended power up sequence: $V_{SS} \rightarrow V_{DD} \rightarrow V_{EE} \rightarrow$ Input Signals $\rightarrow V_{GG}$.
Recommended power down sequence: reverse the above power up sequence.
Avoid floating of the logic power while and are applied.

Recommended operating conditions (referenced to V_{SS} = 0 V)

Item	Min	Typ	Max	Unit
V_{DD}	2.2	3.3	3.6	Volts
V_{GG}	10		35	Volts
V_{EE}	–15		–5	Volts
$V_{GG} - V_{EE}$	15		40	Volts
f_{CLK}			200	kHz
T_A (operating ambient temp)	–25		75	°C

Liquid-Crystal Control Output Voltage



The input signal level CLK, L/R, OE, ONA, SD1 and SD2 swing between V_{DD} and V_{SS}. The signal output level of start pulse (SD1 or SD2) to the next gate driver IC is V_{DD} for "H" and V_{SS} for "L".

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DC Specification

(T_A = –20 to +75°C, V_{DD} = 3.3 V±0.3V, V_{GG} = 25V, V_{EE} = –10V, V_{SS} = 0V)

Item	Description	Condition	Min	Typ	Max	Unit	Remark
V_{IL}	Input LOW voltage		V_{SS}		0.2*V_{DD}	V	All inputs
V_{IH}	Input HIGH voltage		0.8*V_{DD}		V_{DD}	V	All inputs
V_{OL}	Output LOW voltage	I_{OL} = 40mA	V_{SS}		V_{SS}+0.4	V	All I/O's
V_{OH}	Output HIGH voltage	I_{OH} = –40mA	V_{DD}–0.4		V_{DD}	V	All I/O's
R_{OL}	Output LOW resistance	Vout = V_{EE} + 0.5V (*1)			500	Ω	Out 1 to Out 300
R_{OH}	Output HIGH resistance	Vout = V_{GG} – 0.5V (*1)			500	Ω	Out 1 to Out 300
I_{IL}	Input current @LOW	V_{IL} = V_{SS}	–5.0		+5.0	uA	All inputs
I_{IH}	Input current @HIGH	V_{IH} = V_{DD}	–5.0		+5.0	uA	All inputs
I_{DD}	Current from V_{DD}	Note (*2)			500	uA	
I_{EE}	Current from V_{EE}	Note (*2)			-200	uA	
I_{GG}	Current from V_{GG}	Note (*2)			200	uA	

(1) V_{GG} = 25V, V_{EE} = –10V

(2) Current consumption by a 1/768-duty liquid crystal display. No output load, V_{IH} = V_{DD}, V_{IL} = V_{SS}, f_{CLK}=50 kHz, ONA=V_{IH}.

AC Specifications

(T_A = –20 to +75°C, V_{DD} = 3.3 V±0.3V, V_{GG} = 23V, V_{EE} = –10V, V_{SS} = 0V)

Item	Symbol	Condition	Min	Max	Unit
Clock frequency	f_{CLK}			200	kHz
Clock pulse width	t_{CLKH}	50% duty cycle	1		us
Clock pulse width	t_{CLKL}	50% duty cycle	1		us
OE pulse width	t_{OE}		1		us
ONA pulse width	t_{ONA}		3		us
S1A/B and S2A/B set up time	tsu		700		ns
S1A/B and S2A/B hold time	thd		700		ns
Delay time (1)	Tpd1	C_L = 300 pF		1000	ns
Delay time (2)	Tpd2	C_L = 30 pF		800	ns
Delay time (3)	Tpd3	C_L = 300 pF		800	ns
Delay time (4)	Tpd4	C_L = 300 pF		3000	ns

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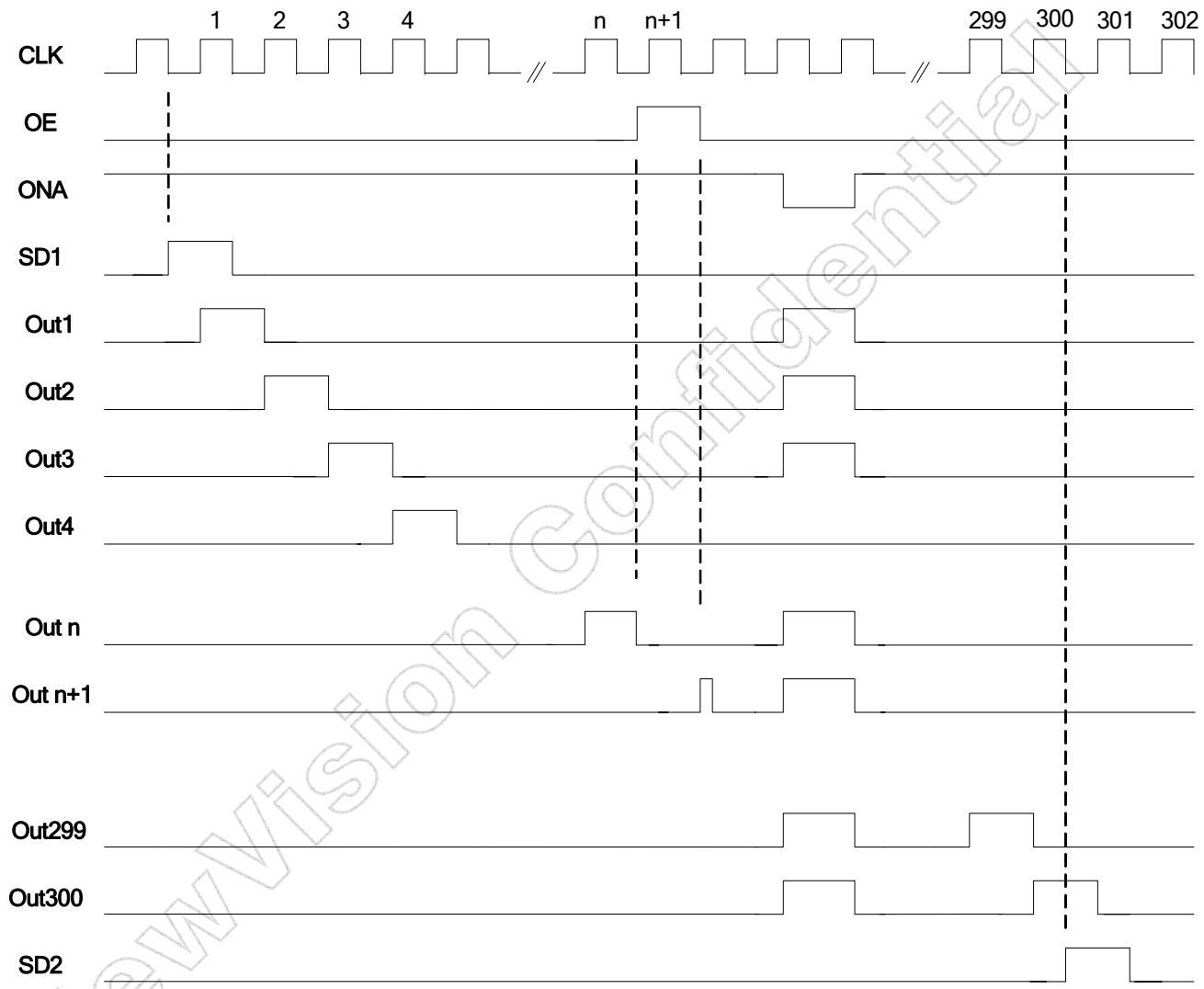
Waveform

Logic Level Description (L/R =”H”, ONA =”H”)



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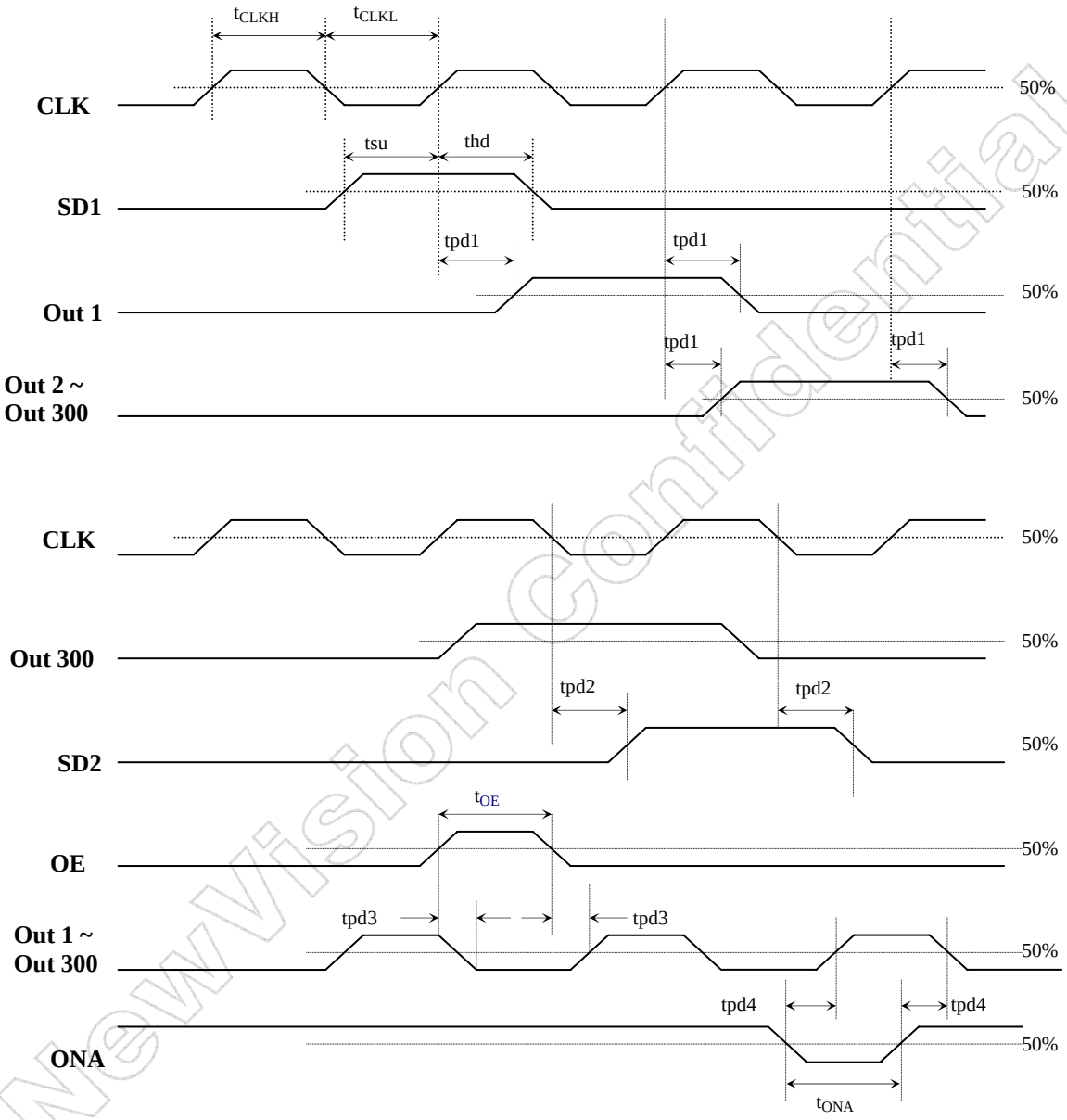
AC Timing Diagram (L/R = "H)



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Notes for CMOS Devices

PRECAUTION AGAINST ESD FOR SEMICONDUCTORS – Strong electric field, when exposed to a MOS device, can cause destruction of the gate oxide and ultimately degrade the device operation. Steps must be taken to stop generation of static electricity as much as possible, and quickly dissipate it once, when it has occurred. Environmental control must be adequate. When it is dry, humidifier should be used. It is recommended to avoid using insulators that easily build static electricity. Semiconductor devices must be stored and transported in an anti-static container, static shielding bag or conductive material. All test and measurement tools including workbench and floor should be grounded. The operator should be grounded using wrist strap. Semiconductor devices must not be touched with bard hands. Similar precautions need to be taken for printed circuit boards with semiconductor devices on it.

HANDLING OF UNUSED INPUT PINS FOR CMOS – No connection for CMOS device inputs can be cause of malfunction. If no connection is provided to the input pins, it is possible that an internal input level may be generated due to noise, etc., hence causing malfunction. CMOS devices behave differently than bipolar or NMOS devices. Input levels of CMOS devices must be fixed high or low by using a pull-up or pull-down circuitry. Each unused pin should be connected to VDD or GND with a resistor, if it is considered to have a possibility of being an output pin. All handling related to the unused pins must be judged device-by-device and related specifications governing the devices.

STATUS BEFORE INITIALIZATION OF MOS DEVICES – Power-on does not necessarily define initial status of MOS device. Production process of MOS does not define the initial operation status of the device. Immediately after the power source is turned ON, the devices with reset function have not yet been initialized. Hence, power-on does not guarantee output pin levels, I/O settings or contents or registers. Device is not initialized until the reset signal is received. Reset operation must be executed immediately after power-on for devices having reset function.

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